20 20 split keyboard



Features:

Hotswap
Wireless: Bluetooth support via ZMK.
Customizable: via ZMK Studio, at zmk.studio

Layout: Plank-ish 40% when magnetically attached, ortholinear 24 + 24 when split

Build Video: YouTube
Firmware and more details: GitHub

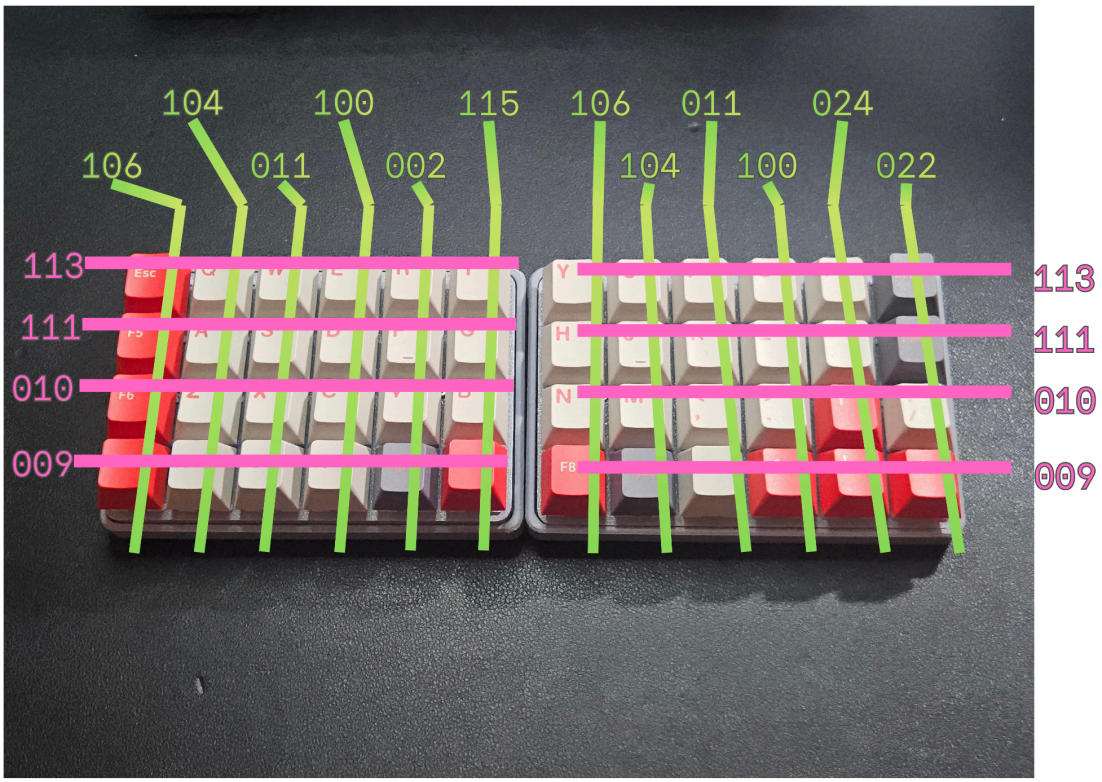
Pinout

Left Half

Columns:	Rows:
0 - 106 1 - 104 2 - 011 3 - 100 4 - 024 5 - 022	0 - 113 1 - 111 2 - 010 3 - 009

Right Half

Columns:	Rows:
0 - 106 1 - 104 2 - 011 3 - 100 4 - 024 5 - 022	0 - 113 1 - 111 2 - 010 3 - 009



If you want to use prebuild firmware, you have to use provided pinout for the matrix

Bill of materials

Assuming you have solder, solder iron, and some wires
Additionally you will need some rubber feet, heatshrinks, hot glue and superglue

Item	Link	Amount
Mechanical Cherry switches	Link	48
Pro micro NRF52840 controller (Nice!Nano v2 clone)	Link	2
Hot Swap Sockets For Mx Cherry	Link	48
Diodes 1N4148	Link	48
M2, 10 mm Socket Head Cap Machine Screws (SHCS)	Link	4
M2, 4 mm Socket Head Cap Machine Screws (SHCS)	Link	4
M2, 6mm stand offs	Link	4
4 by 2 mm magnets	Link	6
Kapton tape	Link	1
Copper wire	Link	1
801350 3.7V Li-Po Battery	Link	2
Slide switch	Link	2
EVA foam (optional but recommended)	Link	
JST battery connectors (optional but recommended)	Link	

Build instructions

For Wiring process reference Build Video: YouTube

Step 1: Prepare everything (Printed part and materials)

Step 2: Put Hotswap sockets in the hotswap plate

Step 3: Put together hotswap plate and switch plate, screw them using m2x10 screws and standoffs

Step 4: Insert the switches

Step 5: Insert and solder the diodes to the hotswap sockets, placing diodes in their place in the hotswap plate

Step 6: Wire other legs of the diodes to create rows

Step 7: Isolate the rows using kapton tape

Step 8: Cut the copper wire for the columns and wire it to other hotswap contact

Step 9: Wire rows and columns to the controller

Step 10: Solder on/off switch with battery and the controller

Step 11: Flash the firmware to the board and test it (Remember that right half wont work if the left is not running)

Step 12: Insert controller into the case, add some hot glue to hold it into the place, insert switch in its socket and screw the bottom of the case to the plate using m2x4 screws

Step 13: Add keycaps and glue the magnets to the case

Step 14: Repeat the same for the other half and enjoy your new keyboard